Bryan A. Pineda

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EDUCATION

Stanford University Stanford, CA
Bachelor of Science Expected June 2026
Major in Computer Science, Artificial Intelligence Track

WORK EXPERIENCE

Alterion Inc. San Francisco, CA
Artificial Intelligence Engineer Jan 2026 - Present

- Led a team of AI engineers to architect an advanced LLM infrastructure, engineering a GNN based orchestration MOE layer and multi-tier memory to power dynamic tool retrieval and execution self-correction.
- Built a high-throughput TDM leak detection pipeline, hydrating data from Supabase SQL to ClickHouse; developed a hybrid detection engine (GBDT and fine-tuned mDeBERTa) outperforming the STDD-in-the-wild baseline by +0.031 and +0.045 AUC without end-to-end supervised training.
- Architected an mBERT LLM guard model for violation detection (Prompt Injection, PII, Toxicity, SQLi), achieving 0.993 AUPRC on deepset and 97.9% accuracy on NotInject, outperforming ProtectAI v2 and Sentinel, resulting in an authored technical paper.
- Built a GitHub Actions CI pipeline with intelligent layer caching and automated vulnerability scanning on PRs, reducing average image build time by 60% (15m to 6m).

Software Engineer Intern Sep 2025 - Jan 2026

- Built an LLM-powered Slack bot (LangChain + Gemini) for personalized recommendations and shipped V1 of an audit logging system for user actions, timestamps, and locations.

PROJECTS & RESEARCH

END-TO-END TRAJECTORY PREDICTION FOR AUTONOMOUS DRIVING Apr 2025 - Jul 2025

- Developed a Transformer-based model for 5-second autonomous-vehicle trajectory prediction using Waymo and nuScenes data, extending TransFuser with BEV prediction and sequential outputs.
- Achieved stable convergence with low prediction error on real-world driving scenarios

AUTONOMOUS DRIVING SIMULATOR Apr 2026 - Present

- Designing a modular Python simulator with components for simulation, perception, planning, control, and a conversational decision-explainer agent.
- Established project foundation through architecture design, implementation planning, and development standards for testing/linting (pytest, ruff); currently building the MVP simulation loop.

QUERY-AWARE MULTI-MODAL GNN RECOMMENDATION SYSTEM Oct 2025 - Dec 2025

- Built a recommendation engine fusing conversational retrieval concepts with Graph Neural Networks to return movies from natural language queries.
- Constructed a unified heterogeneous graph from MovieLens, INSPIRED2, and ReDial; used LLM-generated embeddings and FAISS for efficient candidate generation.

ADDITIONAL

Languages: Python, C++, SQL, JavaScript, TypeScript, Rust
Frameworks/Libraries: PyTorch, TensorFlow, React, Node.js, Spark, Pandas, NumPy
Tools/Platforms: Git, Docker, Kubernetes, GitHub Actions, k6, AWS, GCP, Linux